mdstats legacy/manual Stage 11E-M2 trajectory site-assignment specification

mdstats

2026-07-24 (documentation reclassification only)

Contents

Scope

Legacy/manual Stage 11E-M2 maps persistent Stage-11E-M1 geometric site hypotheses onto mobile-ion trajectory coordinates using the instantaneous Stage-11B/11C2 tile and ring frames. The result is descriptive geometric evidence. It does not certify a metastable free-energy basin, a barrier, or a Markov model.

The public entry point is:

```
assign_trajectory_sites(
    collection,
    frame_tiling_geometry,
    frame_ring_geometry,
    site_topology,
    site_network,
    assignment_profile,
    *,
    species=None,
    atom_indices=None,
    options=None,
    resources=None,
)
```

The analysis requires a trajectory and one explicit assignment profile bound to the Stage-11E-M1 topology profile. This implemented supplied-model branch is not the new data-driven Stage 11E2 deterministic-attractor stage. Public module names and release behavior are unchanged by this documentation reclassification.

Source identity and frame alignment

The implementation shall reject inputs unless:

- the frame-ring catalog is bound to the supplied frame-tiling catalog;
- both frame catalogs use the same selected collection-frame sequence;
- the Stage-11E-M1 topology is bound to the same reference tile and ring catalogs;
- the structural network is bound to the supplied site topology;
- the selected atoms all match the declared target species; and
- the selected frames are strictly increasing trajectory frames.

No frame, ring, site, or ion identity is re-enumerated.

Explicit basin rules

A `SiteAssignmentRule` selects states by an exact state key or by a conjunction of state kind, optional landscape regime, and optional interface label. Every Stage-11E1 state must match exactly one rule.

For a ring-local point state with expected local coordinate (z_0, u_0, v_0) and instantaneous ion coordinate (z, u, v), define

$$S^2 = \left(\frac{z - z_0}{a_z}\right)^2 + \frac{(u - u_0)^2 + (v - v_0)^2}{a_\perp^2}.$$

For a cage-interior state, use the isotropic score

$$S = \frac{\|\Delta \mathbf{r}\|}{a_\perp}.$$

For an annular state of radius ρ_0,

$$S^2 = \left(\frac{z}{a_z}\right)^2 + \left(\frac{\rho - \rho_0}{a_\perp}\right)^2, \qquad \theta = \operatorname{atan2}(v, u).$$

The core basin is $S \le 1$. The explicit transition shell is

$$1 < S \le \gamma,$$

where `transition_multiplier = gamma > 1`.

There is no automatic nearest-state fallback.

Assignment outcomes

For each selected ion and frame:

1. exactly one core candidate gives `assigned`, or `annular_assigned` for an annular state;
2. two or more core candidates give `ambiguous`;
3. no core candidate but one or more transition-shell candidates gives `transition_region`;
4. no accepted or transition candidate gives `unassigned`; and
5. if no state geometry can be realized in the frame, the outcome is `frame_unresolved`.

Candidate diagnostics are deterministically ordered by core membership, transition membership, score, state index, and image shift. The result retains only a configured finite number of leading diagnostics.

Periodic image convention

For every candidate, the general-cell minimum-image operation is applied to the raw site-to-ion Cartesian displacement. If

$$\Delta \mathbf{r}_{\mathrm{MIC}} = \Delta \mathbf{r}_{\mathrm{raw}} + \mathbf{m} H,$$

then the closest site image is labelled by

$$\mathbf{s} = -\mathbf{m}.$$

An observed accepted transition from $(i, \mathbf{s}_i)$ to $(j, \mathbf{s}_j)$ carries

$$\lambda_{ij}^{\mathrm{obs}} = \mathbf{s}_j - \mathbf{s}_i.$$

A structural network match requires the source state, target state, and periodic translation to agree exactly with a Stage-11E1 candidate edge. Multiple matching multigraph edges are retained. Events with no matching edge are reported explicitly as off-network observations.

Temporal statistics

Each ion receives a dense frame-state sequence. Physical state IDs occupy `0..n_physical_states-1`; four deterministic auxiliary IDs represent:

- ambiguous;
- transition region;
- unassigned; and
- frame unresolved.

The complete sequence is passed to `compute_state_transition_statistics()`. Physical occupancy arrays and accepted state-to-state events are also reported separately. A boundary involving an auxiliary state is not converted into a direct physical transition across the gap.

A change in periodic image label is an observed event even when the physical state index is unchanged.

Immutability, resources, and replay

All arrays are defensive read-only copies and all nested metadata are immutable. Resource preflight bounds frames, selected ions, states, candidate evaluations, and retained diagnostics before assignment begins.

Canonical serialization is source-bound. `from_dict()` reruns the calculation with the supplied sources and rejects any noncanonical or tampered payload.

Deferred boundaries

Stage 11E2 does not provide:

- energetic basin certification;
- probabilistic soft assignment;
- hidden-state inference or temporal smoothing;
- transition-state or rate estimation;
- censoring-corrected survival analysis;
- many-ion occupancy constraints; or
- automatic basin-width inference from ring order or ionic radius.